

Donghyun Kwon

ASSOCIATE PROFESSOR

School of Computer Science and Engineering
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Education

Seoul National University

PH.D IN DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

- Thesis: Hardware-assisted Isolation for System Framework to Ensure Kernel Integrity
- Advisor: Yunheung Paek
- Distinguished Ph.D. Dissertation Award

Seoul, Korea

Mar 2012 - Feb 2019

Seoul National University

B.S IN DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Seoul, Korea

Mar 2008 - Feb 2012

Professional Experience

Associate Professor

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING, PUSAN NATIONAL UNIVERSITY

Pusan, South Korea

Mar 2023 - Present

Assistant Professor

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING, PUSAN NATIONAL UNIVERSITY

Pusan, South Korea

Mar 2020 - Feb 2023

Senior Researcher

ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE

Daejeon, South Korea

Mar 2019 - Feb 2020

Publications

* *corresponding author*

1. **SERA: Secure Micro XRCE-DDS Establishment with Remote Attestation for micro-ROS**
Jeonghwan Kang, Seungmin Kang, Euijun Kim, Jaeseon Park, Kyoungwan Kim, and **Donghyun Kwon***
IEEE Access, 2026
2. **WARD: Efficient Memory Protection for WebAssembly on Tiny Embedded Systems**
Chaewon Shin, Gowon Han, Suhyeon Song, Jinsun Park, Yoon-ho Choi, and **Donghyun Kwon***
IEEE Access, 2026
3. **Beyond Address Spaces: In-process Memory Isolation for RISC-V**
Seonghwan Park, Hayoung Kang, and **Donghyun Kwon***
Computers & Security, 2026
4. **SMORE: Practical Redzone-Based Stack Memory Error Detection Mechanism for Embedded Systems**
Jaeyeol Park, Yunju Gu, and **Donghyun Kwon***
Annual Computer Security Applications Conference (ACSAC), 2025
5. **ELK: Effective Lock-and-Key Technique for Temporal Memory Safety on Embedded Devices in ARMv8-M**
Jeonghwan Kang, Kyoungwan Kim, and **Donghyun Kwon***
Annual Computer Security Applications Conference (ACSAC), 2025
6. **Watch Your Callback: Offline Anomaly Detection using Machine Learning in ROS 2**
Jeonghwan Kang, Kyoungwan Kim, and **Donghyun Kwon***
IEEE Access, 2025

7. **WasDom: An Efficient Write Protection for Wasm JITed Code With ARM Domain**
Suhyeon Song, Chaewon Shin, and **Donghyun Kwon***
IEEE Access, 2025
8. **ROSec: Intra-Process Isolation for ROS Composition with Memory Protection Keys**
Jiwon Seo, Kayondo Martin, Jeonghwan Kang, **Donghyun Kwon***, and Yunheung Paek*
IEEE Transactions on Automation Science and Engineering (T-ASE), 2025
9. **Non-volatile flip-flop with soft error tolerant capability using DICE and C-element**
Shogo Takahashi, **Donghyun Kwon**, and Kazuteru Namba*
IEICE Nonlinear Theory and Its Applications, 2024
10. **Look Before You Access: Efficient Heap Memory Safety for Embedded Systems on ARMv8-M**
Jeonghwan Kang, Jaeyeol Park, Jiwon Seo, and **Donghyun Kwon***
Design Automation Conference (DAC), 2024
11. **Enhancing ad Lock-and-key Scheme with MTE to Mitigate Use-After-Frees**
Inyoung Bang, Martin Kayondo, Junseung Yu, **Donghyun Kwon**, Yeongpil Cho*, and Yunheung Paek*
IEEE Access, 2023
12. **metaSafer: A Technique to Detect Heap Metadata Corruption in WebAssembly**
Suhyeon Song, Seonghwan Park, and **Donghyun Kwon***
IEEE Access, 2023
13. **ZOMETAG: Zone-based Memory Tagging for Fast, Deterministic Detection of Spatial Memory Violations on ARM**
Jiwon Seo, Junseung You, **Donghyun Kwon**, Yeongpil Cho*, and Yunheung Paek*
IEEE Transactions on Information Forensics and Security (TIFS), 2023
14. **SFITAG: Efficient Software Fault Isolation with Memory Tagging for ARM Kernel Extensions**
Jiwon Seo, Junseung You, Yungi Cho, Yeongpil Cho, **Donghyun Kwon***, Yunheung Paek*
ACM Asia Computers and Communications Security (ASIACCS), 2023
15. **DEMIX: Domain-Enforced Memory Isolation for Embedded System**
Haeyoung Kim, Harashta Tatimma Larasati, Jonguk Park, Howon Kim, and **Donghyun Kwon***
Sensors, 2023
16. **Exploring Effective Uses of the Tagged Memory for Reducing Bounds Checking Overheads**
Jiwon Seo, Inyoung Bang, Yungi Cho, Jangseop Shin, Dongil Hwang, **Donghyun Kwon**, Yeongpill Cho, Yunheung Paek
The Journal of Supercomputing, 2022
17. **Efficient CFI Enforcement for Embedded Systems Using ARM TrustZone-M**
Gisu Yeo, Yeryeong Kim, Suhyeon Song, **Donghyun Kwon***
IEEE Access, 2022
18. **Bratter: An instruction set extension for forward control-flow integrity in RISC-V**
Seonghwan Park, Dongwook Kang, Jeonghwan Kang, **Donghyun Kwon***
Sensors, 2022
19. **CoMeT: Configurable Tagged Memory Extension**
Jinjaee Lee, Derry Pratama, Minjae Kim, Howon Kim, **Donghyun Kwon***
Sensors, 2021
20. **A Hardware Platform for Ensuring OS Kernel Integrity on RISC-V**
Donghyun Kwon, Dongil Hwang, Yunheung Paek
Electronics, 2021
21. **RIMI: Instruction-level Memory Isolation form Embedded Systems on RISC-V**
Haeyoung Kim, Jinjaee Lee, Derry Pratama, Asep Muhamad Awaludin, Howon Kim, **Donghyun Kwon***
International Conference On Computer Aided Design (ICCAD), 2020
22. **PrOS: Light-weight Privatized Secure OSeS in ARM TrustZone**
Donghyun Kwon, Jiwon Seo, Yeongpil Cho, Byoungyoung Lee, Yunheung Paek
IEEE Transactions on Mobile Computing (TMC), 2020
23. **RiskIM: Toward Complete Kernel Protection with Hardware Support**
Dongil Hwang, Myunghoon Yang, Seongil Jeon, Younghun Lee, **Donghyun Kwon***, Yunheung Paek
Design, Automation and Test in Europe (DATE), 2019

24. **Safe and Efficient Implementation of a Security System on ARM using Intra-Level Privilege Separation**
Donghyun Kwon, Hayoon Yi, Yeongpil Cho, Yunheung Paek
ACM Transactions on Privacy and Security (TOPS), 2019
25. **CRCCount: Pointer Invalidation with Reference Counting on Mitigate Use-After-Free in Legacy C/C++**
Jangseop Shin, Donghyun Kwon, Jiwon Seo, Yeongpil Cho, Yunheung Paek
Network and Distributed System Security Symposium (NDSS), 2019
26. **uXOM: Efficient eXecute-Only Memory on ARM Cortex-M**
Donghyun Kwon, Jangseop Shin, Giyeol Kim, Byoungyoung Lee, Yeongpil Cho, Yunheung Paek
USENIX Security, 2019
27. **Hypernel: A Hardware-Assisted Framework for Kernel Protection without Nested Paging**
Donghyun Kwon**, Kuenwhee Oh**, Junmo Park, Seungyong Yang, Yeongpil Cho, Brent Byunghoon Kang, Yunheung Paek
Design Automation Conference (DAC), 2018
(**) joint first authors
28. **VM-CFI: Control-Flow Integrity for Virtual Machine Kernel Using Intel PT**
Donghyun Kwon, Jiwon Seo, Sehyun Baek, Giyeol Kim, Sunwoo Ahn, Yunheung Paek
International Conference on Computational Science and Its Applications (ICCSA), 2018
29. **Instruction-Level Data Isolation for the Kernel on ARM**
Yeongpil Cho, Donghyun Kwon, Yunheung Paek
Design Automation Conference (DAC), 2017
30. **Optimization Techniques to Enable Execution Offloading for 3D Video Games**
Donghyun Kwon, Seungjun Yang, Yunheung Paek, Kwangman Ko
Multimedia Tools and Applications (MTAP), 2017
31. **Dynamic Virtual Address Range Adjustment for Intra-Level Privilege Separation on ARM**
Yeongpil Cho, Donghyun Kwon, Hayoon Yi, Yunheung Paek
Network and Distributed System Security Symposium (NDSS), 2017
32. **Precise Execution Offloading for Applications with Dynamic Behavior in Mobile Cloud Computing**
Yongin Kwon, Hayoon Yi, Donghyun Kwon, Seungjun Yang, Yeongpil Cho, Yunheung Paek
Pervasive and Mobile Computing (PMC), 2016
33. **Hardware-Assisted On-Demand Hypervisor Activation for Efficient Security Critical Code Execution on Mobile Devices**
Yeongpil Cho, Jun-Bum Shin, Donghyun Kwon, MyungJoo Ham, Yuna Kim, Yunheung Paek
USENIX Annual Technical Conference (ATC), 2016
34. **Mantis: Efficient Predictions of Execution Time, Energy Usage, Memory Usage and Network Usage on Smart Mobile Devices**
Yongin Kwon, Sangmin Lee, Hayoon Yi, Donghyun Kwon, Seungjun Yang, Byung-gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, Yunheung Paek
IEEE Transactions on Mobile Computing (TMC), 2015
35. **Techniques to Minimize State Transfer Costs for Dynamic Execution Offloading in Mobile Cloud Computing**
Seungjun Yang, Donghyun Kwon, Hayoon Yi, Yeongpil Cho, Yongin Kwon, Yunheung Paek
IEEE Transactions on Mobile Computing (TMC), 2014
36. **Fast Dynamic Execution Offloading for Efficient Mobile Cloud Computing**
Seungjun Yang, Yongin Kwon, Yeongpil Cho, Hayoon Yi, Donghyun Kwon, Jonghee Youn, Yunheung Paek
IEEE International Conference on Pervasive Computing and Communications (PerCom), 2013
37. **Mantis: Automatic Performance Prediction for Smartphone Applications**
Yongin Kwon, Sangmin Lee, Hayoon Yi, Donghyun Kwon, Seungjun Yang, Byung-Gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, Yunheung Paek
USENIX Annual Technical Conference (ATC), 2013

Academic Service

- (Reviewer) ACM Transactions on Privacy and Security (TOPS)
- (Reviewer) ACM Transactions on Architecture and Code Optimization (TACO)

- (Reviewer) IEEE Transactions on Information Forensics & Security (TIFS)
- (Reviewer) IEEE Transactions on Computers (TC)
- (Reviewer) IEEE Access
- (Reviewer) Journal of Supercomputing
- (Associate Editor) Journal of Information Processing Systems (JIPS)

Teaching

- Grad. **System Vulnerabilities and Countermeasures**, Fall 2021 - Fall 2025
- Grad. **Introduction to System and Software Security**, Spring 2021-Spring 2025
- Under. **Web Application Programming**, Fall 2021 - Fall 2023
- Under. **Logic Circuit Design & LAB.**, Fall 2023 - Fall 2024
- Under. **Compiler**, Fall 2022
- Under. **Computer Architecture**, Fall 2024 - Fall 2025
- Under. **Blockchain**, Spring 2024 - Spring 2025
- Under. **Programming Principles and Practice**, Fall 2020 - Fall 2021
- Under. **Adventure Design**, Fall 2020
- Under. **Introduction to Internet and Web**, Spring 2020 - Spring 2023
- Under. **System Software**, Spring 2020 - Spring 2025
- Under. **Discrete Mathematics 1**, Spring 2020